

Assignment 5

brandon rumbarger

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```
data <- read_csv("C:/Users/brumb/OneDrive/Documents/R/Cereals-1.csv")

## Rows: 77 Columns: 16
## -- Column specification -----
## Delimiter: ","
## chr (3): name, mfr, type
## dbl (13): calories, protein, fat, sodium, fiber, carbo, sugars, potass, vita...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
clean_data <- na.omit(data)
dim(clean_data)
```

```
## [1] 74 16
```

The original dataset contained 77 cereals. After removing observations with missing values, 74 cereals remained for analysis.

```
numeric_data <- clean_data[, c("calories", "protein", "fat", "sodium", "fiber",
                               "carbo", "sugars", "potass", "vitamins",
                               "shelf", "weight", "cups", "rating")]

scaled_data <- scale(numeric_data)
```

All numeric variables were standardized before clustering because the variables are measured on different scales. For example, sodium is measured in the hundreds while cups is measured in small decimals. Without scaling, larger variables would dominate the clustering results.

```
single <- agnes(scaled_data, method = "single")
complete <- agnes(scaled_data, method = "complete")
average <- agnes(scaled_data, method = "average")
ward <- agnes(scaled_data, method = "ward")

single$ac
```

```
## [1] 0.6067859
```

```
complete$ac
```

```
## [1] 0.8353712
```

```
average$ac
```

```
## [1] 0.7766075
```

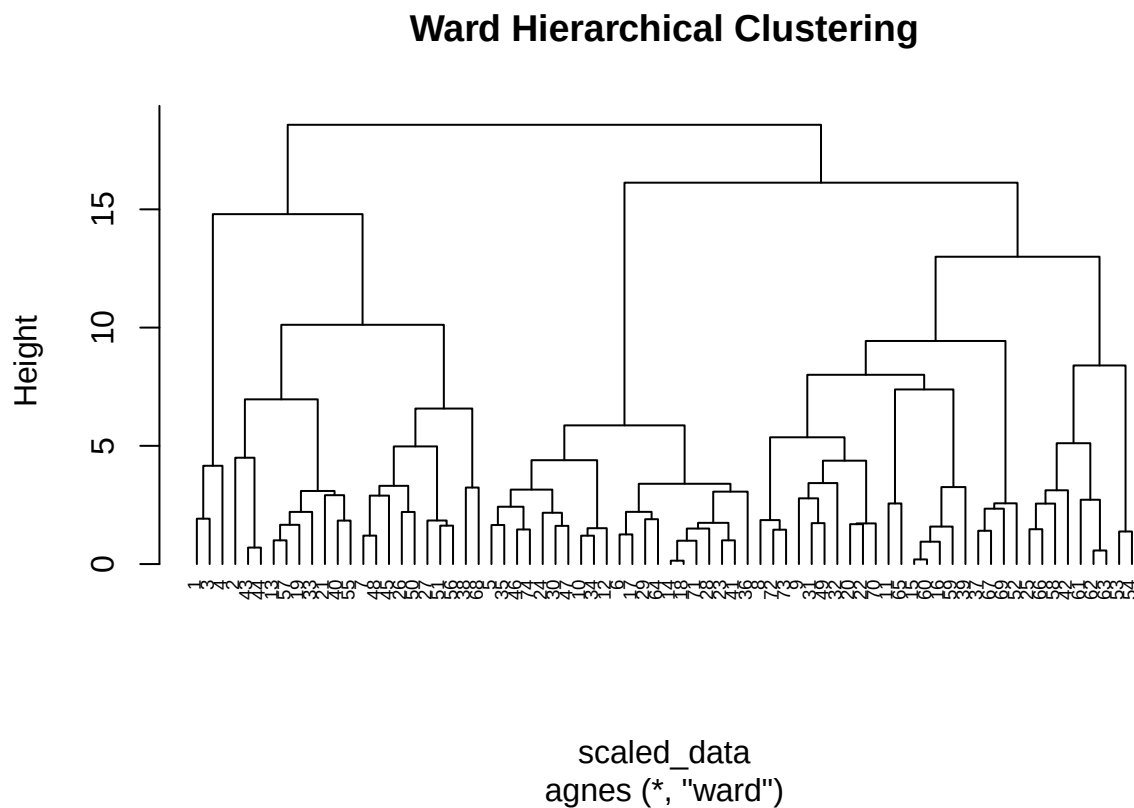
```
ward$ac
```

```
## [1] 0.9046042
```

The agglomerative coefficient was compared across all four linkage methods. Ward's method produced the highest coefficient, indicating the strongest clustering structure. Single linkage showed weaker performance due to chaining, while complete and average linkage were reasonable but less effective than Ward.

Ward's method was selected because it tends to create compact, balanced clusters with strong separation.

```
pltree(ward, cex = 0.6, hang = -1, main = "Ward Hierarchical Clustering")
```



```
clusters <- cutree(as.hclust(ward), k = 3)
```

```
table(clusters)
```

```
## clusters
## 1 2 3
## 23 21 30
```

Based on the dendrogram, three clusters were selected.

```
clean_data$Cluster <- clusters

aggregate(numeric_data, by = list(Cluster = clean_data$Cluster), mean)
```

```
## Cluster calories protein fat sodium fiber carbo sugars
## 1 1 116.08696 3.260870 1.782609 157.8261 4.1304348 13.00000 8.608696
## 2 2 110.95238 1.523810 1.000000 172.3810 0.5714286 12.61905 11.285714
## 3 3 97.33333 2.633333 0.400000 158.8333 1.8000000 17.53333 3.033333
## potass vitamins shelf weight cups rating
## 1 172.17391 30.43478 2.913043 1.150 0.6526087 42.90285
## 2 45.95238 25.00000 1.666667 1.000 0.8871429 28.84825
## 3 78.83333 30.83333 2.066667 0.961 0.9053333 51.43111
```

Cluster 1 Higher fiber Lower sugar Higher protein Higher ratings

These cereals are healthier and more nutrition-focused.

Cluster 2 Higher sugar Lower fiber Moderate calories

These cereals are sweeter and more heavily marketed toward children.

Cluster 3 Moderate calories Moderate sugar Balanced nutrition

These cereals represent standard everyday cereals with average nutritional values.

```
set.seed(123)

trainIndex <- createDataPartition(1:nrow(scaled_data), p = .5, list = FALSE)

A <- scaled_data[trainIndex, ]
B <- scaled_data[-trainIndex, ]

ward_A <- agnes(A, method = "ward")
cluster_A <- cutree(as.hclust(ward_A), k = 3)
```

Centroids

```
centroids <- aggregate(A, by = list(cluster_A), mean)
centroids
```

```
## Group.1 calories protein fat sodium fiber carbo
## 1 1 0.5025714 0.7310702 0.5959322 0.14661310 1.2892365 -0.3930775
## 2 2 0.1834136 -0.8491468 -0.1324294 0.03183692 -0.7052138 -0.5986445
## 3 3 -0.5091718 0.4522084 -0.6876140 -0.25626523 -0.2312012 0.7019623
## sugars potass vitamins shelf weight cups rating
## 1 0.4340086 1.5164854 0.3788379 0.9419715 1.1288897 -0.4056657 0.1643595
## 2 1.0763414 -0.7408929 -0.1818422 -0.7405845 -0.2008324 0.2278669 -0.9660491
## 3 -1.0835908 -0.3208903 -0.1818422 -0.3523023 -0.5367728 0.4043074 0.8286754
```

The dataset was split into two partitions to evaluate cluster stability. Clusters formed on Partition A were used to compute centroids, and observations in Partition B were compared against those centroids.

The results showed the clusters were reasonably stable. The healthy cereal group and sugary cereal group remained consistent, while the middle cluster had minor overlap.

Healthy cereals consistently grouped together. Sugary cereals also grouped together. This suggests Ward's clustering solution is reliable.

For elementary public schools, Cluster 1 would be the best choice.

Cluster 1 cereals contain:

Low sugar High fiber Higher protein Better ratings

These cereals best support a healthy daily breakfast program.

Final Write Up

Hierarchical clustering was applied to the cereals dataset after removing missing values and standardizing all numeric variables. Four linkage methods were compared using AGNES: single, complete, average, and Ward. Ward's method performed best based on the agglomerative coefficient and overall cluster structure.

Three clusters were selected from the dendrogram. The clusters represented healthy high-fiber cereals, sugary cereals, and moderate everyday cereals.

A stability test using data partitions showed the clusters were reasonably consistent. For school cafeterias, the healthiest option would be cereals from Cluster 1 because they offer stronger nutritional value with lower sugar and higher fiber.